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ABSTRACT

Image manipulation detection methods play a pivotal role in safeguarding digital image authenticity and integrity by identifying and locating manipulations. Existing image manipulation detection methods suffer from limited generalization, as it is difficult for existing training datasets to cover different manipulation modalities in the open domain. In this paper, we propose a Generalization Enhancement Strategy (GES) based on data augmentation and ensemble learning. Specifically, the GES consists of two modules, namely an Additive Image Manipulation Data Augmentation (AIM-DA) module and a Mask Confidence Estimate based Ensemble Learning (MCE-EL) module. In order to take full advantage of the limited number of real and manipulated images, the AIM-DA module enriches the diversity of the data by generating manipulated traces accumulatively with different kinds of manipulation methods. The MCE-EL module is designed to improve the accuracy of detection in the open domain, which is based on computing and integrating the evaluation of the confidence level of the output masks from different image manipulation detection models. Our proposed GES can be adapted to existing popular image manipulation detection methods. Extensive subjective and objective experimental results show that the detection F1 score can be improved by up to 34.9%, and the localization F1 score can be improved by up to 11.7%, which validates the effectiveness of our method.

1. Introduction

Image forensics becomes critically important with the proliferation of deepfake content. Advances in technology lower the barriers to editing multimedia content, enabling almost anyone to quickly become proficient, with modifications becoming increasingly difficult to detect. Consequently, scientifically identifying the originality and authenticity of images becomes a key issue that needs urgent resolution. Image manipulation [1] detection techniques play a crucial role in image forensics, not only determining the overall authenticity of images but also localizing specific tampered areas. Nevertheless, the current image manipulation detection methodologies exhibit a deficiency in generalization, rendering them less effective for deployment in real-world applications. This challenge stems primarily from the fact that images in the real world may be generated through a myriad of unknown tampering operations or newly emerging tampering techniques, devoid of prior knowledge. In this paper, we develop a generalization

enhancement strategy for open-domain image manipulation detection. 'open-domain' refers to the application of image manipulation detection algorithms to a diverse range of tampering techniques and sources that are not limited to specific datasets. We specify that our detection algorithm does not encompass AI-generated content (AIGC) images. But rather traditional forms of image processing such as image splicing [2], copy-movement [3] and other related techniques.

Existing image manipulation detection algorithms can be categorized into methods based on traditional handcrafted features and those utilizing deep learning techniques. Tampered images often retain artifacts or anomalies, which, under certain perspectives, exhibit characteristics inconsistent with original images. Early image manipulation detection algorithms primarily rely on handcrafted features [4–7] to expose such inconsistencies, thereby localizing tampered areas. Inspired by traditional algorithms, deep learning-based methods also leverage similar underlying clues as tampering features, demonstrating exceptional performance. However, these methods are sensitive to specific

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manipulations, making them only suitable for scenarios involving a single type of tampering, such as image splicing [2], copy-move [3], or inpainting [8].

In recent years, many image manipulation detection models aim to identify multiple tampering operations within a unified framework. Researchers embark on explorations in various aspects such as careful feature selection [9], exploitation of contextual information [10], domain adaption [11], and data augmentation [12]. These methods establish numerous universal image manipulation detection models, adapting end-to-end deep neural networks to various tampering operations. However, the generalization of existing methods is unsatisfactory, mainly for the following three reasons.

- Tampering methods in widely used training datasets are limited and differ greatly from the complex tampered images in real-world scenarios. Early datasets experience a domain shift with the swift advancements in generative AI. Acquiring real-world tampered images is both time-consuming and challenging.
- Existing data augmentation techniques do not fully leverage authentic images in datasets and cannot include multiple types of tampering in a single image, leading to an underestimation of existing model performance, such as deepfake-based subtle texture alterations, multi-modal manipulations that blend image, and complex region-based editing techniques, present significant challenges for detection. These operations are often undetectable by models trained on standard datasets due to their novelty and complexity.
- Researchers mainly focus on refining individual models, creating many with exceptional performance. However, they overlook the benefits of ensembling complementary models to improve overall performance.

In this paper, we introduce a simple yet effective GES for open domain image manipulation detection, which can be easily adapted to current image manipulation detection models. To be specific, we first propose an Additive Image Manipulation Data Augmentation (AIM-DA) module composed of multiple cumulative tampering operations designed to simulate real-world tampering, thereby enriching the dataset with a diverse array of tampering traces. We employ SAM [13] to segment real images within the dataset, enabling authentic images to contribute to the data augmentation process. Then each image undergoes a sequential application of additive tampering generation, fusion, and post-processing, ranging from significant alterations such as splicing and copy-move to more nuanced adjustments like image fusion. By layering multiple tampering techniques, the approach aims to closely mimic the complexity and subtlety of tampered images encountered in real-world contexts. To enhance the accuracy of detection, we design a Mask Confidence Estimate based Ensemble Learning (MCE-EL) module. This module strategically combines the outputs of different models by evaluating the confidence levels of their predicted masks. By leveraging the confidence scores, the MCE-EL module selectively synthesizes the most reliable outputs from each participating algorithm, thereby optimizing the overall detection accuracy and localization precision. In summary, our contributions are as follows:

- Our GES approach significantly boosts existing algorithms' performance and offers easy integration and adaptability across different models.
- We propose AIM-DA, a simple yet effective data augmentation strategy. Unlike prior data augmentation approaches that focus narrowly on specific types of image manipulations. It employs SAM and additive image manipulation to maximize the use of both authentic and tampered images for enriching tampering traces. This method boosts the generalization of existing algorithms by providing a diverse set of training data.

- We develop a MCE-EL module that introduces a novel strategy of mask confidence-based voting to improve the accuracy of detection in the open domain. This module integrates the outcomes of multiple image manipulation detection models for a comprehensive multi-perspective examination of images.

The remainder of this paper is organized as follows: In Section 2, we present an overview of the related works. Then, in Section 3 we describe our GES for image manipulation detection models, including an AIM-DA module and a MCE-EL module. Finally, in Section 4 we assess our proposed approach in Cross-domain image manipulation detection through a set of experiments. Our approach has demonstrated superior robustness against common image degradations such as compression and noise, as well as cross-platform manipulation, outperforming state-of-the-art models even under adverse conditions typically introduced by social media platforms. Lastly, Section 5 concludes our work.

2. Related work

In this part, we briefly discuss three lines of related work and further highlight the differences between our proposed algorithm and the existing works.

2.1. Image manipulation detection

Recent approaches in image manipulation detection have evolved to detect and locate various tampered regions within digital images to maintain their authenticity and integrity. These advancements are driven by the increasing diversity of tampering techniques, leading to the need for image manipulation detection models to detect and localize multiple types of manipulations within a single framework. Consequently, this poses significant challenges in terms of the model's generalization capabilities. To tackle this challenge, researchers investigate various approaches, such as reformulating the problem as anomaly detection [14–16], domain adaptation [11], data augmentation [12], adversarial training [17], multi-scale feature learning [18–20], and semantic-agnostic approaches [21,22]. By implementing these approaches, the goal is to improve the network's generalization across multiple domains without the need for fine-tuning. Our work aims at making a further step in enhancing the generalization of image manipulation detection models and proposes a method named GES. It utilizes AIM-DA module to boost the generalization of individual models and employs MCE-EL module to integrate multiple models, further improving generalization.

2.2. Data augmentation

Data augmentation is a straightforward and cost-effective method to generate samples that bolster the model's ability to generalize. The primary goal of data augmentation is to increase the diversity or quantity of existing training data through various operations, such as flipping, rotating, scaling, cropping, and adding noise, mainly to reduce overfitting and improve model generalization. While widely utilized data augmentation techniques such as Cutout [23], GridMask [24], and RandAugment [25] have proven effective, they primarily enhance input images without addressing masks and labels. For instance, employing Cutout in image manipulation detection datasets can lead to confusion, as the randomly masked areas might be misconstrued as tampered regions, potentially diminishing model performance by not accurately reflecting the true nature of image manipulations. Beyond the conventional data augmentation techniques, Domain Randomization [26], Adversarial Data Augmentation [27] and Data Generation [28] are other effective strategies. Mixup [29] and MixStyle [30] are two common data generation techniques, the former directly mixes input data, while the latter mixes in the feature space. In image manipulation detection, methods utilizing data augmentation aim to enhance the generalization of image manipulation detection algorithms by artificially expanding

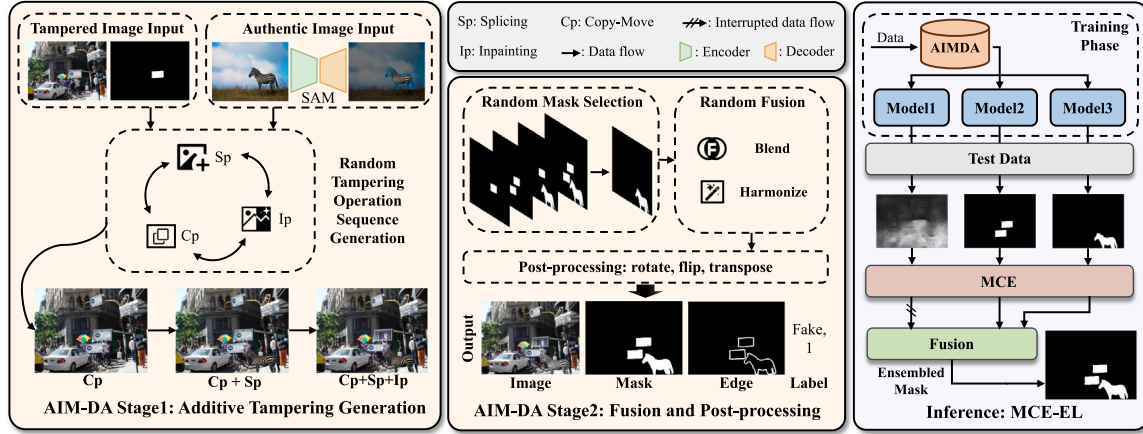


Fig. 1. Framework of our proposed GES. Our GES for image manipulation detection algorithms is divided into two parts: AIM-DA during the training phase and MCE-EL during the inference phase. AIM-DA completes the initial generation of tampered images in the additive tampering generation stage, followed by further refinement of the tampering in the Fusion and Post-processing stage, thus producing a more diverse array of tampering traces. During the inference phase, the MCE-EL is utilized to eliminate low-quality masks and integrate high-quality masks, facilitating model ensemble.

the training dataset with tampered versions of the original images. [31] focuses on detecting Photoshop manipulation traces with a fully convolutional encoder-decoder structure, utilizing Photoshop scripts for realistic training data generation. Zhou et al. [12] adopt a GAN-based approach for generating challenging tampered images. Existing data augmentation approaches often overlook the utilization of the abundant authentic images available in datasets and typically generate tampered images that contain only a singular type of manipulation. In this paper, we present AIM-DA, a method specifically designed for image manipulation detection tasks. AIM-DA enhances existing tampered images in an additive manner while leveraging previously overlooked real images to generate new tampered samples.

2.3. Ensemble learning

Ensemble learning aims to enhance model capabilities through the combination of multiple models, utilizing specific network architecture designs and training strategies to explore relationships across different source domains for improved performance. The underlying assumption of ensemble learning is that any sample can be considered a composite from multiple source domains, making the overall prediction a composite of networks trained across various domains [32]. The core idea is to improve the accuracy, generalization, and robustness by aggregating predictions from multiple models. Methods for achieving this include combining traditional classifiers [33,34] or deep neural networks trained on multiple domains [35], employing fusion strategies to merge different model predictions [36,37], and averaging model weights during training [38,39]. However, a potential limitation of ensemble learning is its increased demand for computational resources, as training and storing multiple models require more computational power and storage space. While ensemble learning has shown promise in classification tasks, its effectiveness in segmentation or localization tasks remains unproven. Further exploration is needed for its application in image manipulation detection tasks. To address this gap, this paper introduces the MCE-EL module. This module filters high-quality predictions for integration, offering an enhancement during the inference phase that eliminates the need for model retraining. This approach is straightforward to implement and can enhance the performance of existing image manipulation detection models.

3. The proposed method

GES is designed considering both enriching data diversity and enhancing discriminative confidence through the AIM-DA module and the MCE-EL, respectively. Fig. 1 illustrates the overall framework of the proposed GES.

3.1. Additive image manipulation data augmentation

In this section, we devise the AIM-DA method. The method consists of two main phases, additive tampering generation and image fusion & post-processing. The additive tampering generation stage is devised to target three prevalent tampering techniques, including splicing, copy-move, and inpainting. This stage leverages both tampered and authentic images within the dataset to generate rich tampering traces. Image fusion can enhance the authenticity of the resulting tampered image. Simple post-processing yields additional samples while preventing obscuring the created tampering traces. Traditional augmentation techniques, such as rotation or Gaussian noise, tend to modify existing images without introducing new types of tampering artifacts. In contrast, AIM-DA actively creates diverse manipulation scenarios that help models learn to detect specific tampering characteristics, thus improving their overall robustness and generalization. We will describe our approach in detail below.

3.1.1. Additive tampering generation

We first preprocess the genuine images in the dataset, followed by the introduction of three distinct data augmentation algorithms. These algorithms contribute to the dataset's complexity and variety, enabling the model to more effectively learn the characteristics of image tampering. This not only speeds up model convergence but also enhances the accuracy and generalization of tampering detection.

Due to the lack of masks for the genuine images in the tampered image dataset, it is necessary to segment the authentic images to obtain tampering masks for each object, facilitating subsequent operations. In terms of selecting a model for image segmentation, we choose the Segment Anything Model (SAM) [13]. We set different parameters for low-resolution (long side less than 512 pixels) images and high-resolution (long side more than 512 pixels) images to get a better segmentation result. After processing with SAM, both the tampered and authentic images are now accompanied by masks. The subsequent step involves the Random Tampering Operation Sequence Generation. A sequence of tampering operations is formed by randomly selecting one or more operations from splicing, copy-move, and inpainting. Following this sequence, additive tampering operations are applied to the image. The algorithms for these three types of tampering operation data augmentation are detailed below.

Image splicing data augmentation: The image splicing data augmentation algorithm involves a series of steps applied to a donor image I_1 , a target image I_2 , and their respective masks M_1 and M_2 . The process will finally generate a spliced tampered image I_{sp} and its corresponding data-augmented masks M_{sp} and E_{sp} .

- **Mask Preparation.** If I_2 is tampered, M_2 is used directly. If I_2 is genuine, M_2 is an all-zero mask matching the size of I_2 . The donor image I_1 and mask M_1 are randomly selected from the training set. If I_1 is tampered, M_1 is its corresponding mask. If I_1 is genuine, one of the SAM-segmented masks associated with it is randomly chosen as M_1 .
- **Tampering Process.** The donor region R_1 is cropped from I_1 using the mask M_1 . R_1 undergoes random resizing, chosen from the following methods: Nearest-neighbor interpolation, Area resize, Bilinear interpolation, Lanczos interpolation, and Bicubic interpolation. After random rotation and brightness-contrast adjustment, R_1 is pasted onto I_2 to generate the augmented spliced tampered image I_{sp} . The masks M_1 and M_2 undergo the same process to result in the data-augmented masks M_{sp} and E_{sp} .

Copy-move data augmentation. The copy-move data augmentation enhances the generalization of manipulation detection models by introducing artificial tampering into authentic images. This method aims to generate a new copy-move image I_{cp} along with its masks M_{cp} and E_{cp} . The process begins with an input image I_1 .

- **Mask Preparation.** Depending on the authenticity of I_1 , the mask M_1 is determined as follows: If I_1 is tampered, its corresponding mask is used as M_1 . If I_1 is authentic, M_1 is randomly selected from the masks segmented by SAM. A copy-move tampering mask is then chosen from either randomly generated masks or M_1 . The randomly generated masks can be rectangular or circular. These are designed to create simple samples to quickly teach the network basic tampering characteristics.
- **Tampering Process.** Similar to the splicing augmentation process, the tampered region undergoes: random scaling, random rotation, and brightness-contrast adjustments. After these operations, the region is pasted onto a random location in I_1 to produce a new copy-move tampered image I_{cp} along with its corresponding mask M_{cp} and E_{cp} .

Inpainting data augmentation. Given the lack of inpainting images in most image manipulation datasets, inpainting data augmentation is crucial. The process initiates with an input image selection: for fake images, the corresponding mask is used. For real images, a mask is randomly selected from those segmented by SAM.

- **Mask Preparation** The selection of the inpainting mask is diversified into three scenarios to mimic real-world tampering more closely. Generating random rectangular or circular masks. Drawing irregular masks from the irregular mask dataset [40] to simulate more complex inpainting scenarios. Directly using the input mask as the inpainting mask.
- **Tampering Process:** The inpainting operation is performed within the area defined by the inpainting mask using both traditional and deep learning-based methods:
 1. **Traditional Methods:** Techniques like Telea [41] and Navier-Stokes [42] algorithms are employed for simple and fast inpainting tasks.
 2. **Deep Learning Methods:** Advanced model GC [43] is utilized to generate high-quality and contextually coherent inpainted regions. This model present a generative image inpainting system to complete images with free-form mask and guidance. The system is based on gated convolutions learned from millions of images without additional labeling efforts.

To increase diversity in the augmented dataset, both traditional and deep learning-based inpainting methods are employed, with a higher probability of employing deep learning approaches during

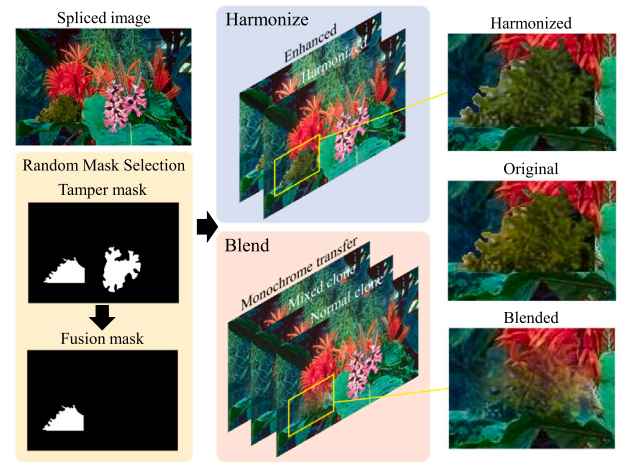


Fig. 2. Random fusion samples. The input image exhibits two tampering areas after splicing data augmentation. A random mask is selected as the fusion mask. The harmonized image undergoes adjustments only within the fusion mask, while the enhanced image is adjusted as a whole. In the blending process, three modes – normal clone, mixed clone, and monochrome transfer – are randomly selected and applied to the fusion mask area. The original image in the rightmost column is cropped from the spliced image.

data augmentation due to their convenience and widespread use. Finally, the output includes the inpainted image I_{ip} , tampered area mask M_{ip} , and edge mask E_{ip} .

3.1.2. Image fusion and post-processing

In real image editing processes, especially for copy-move and splicing manipulations, image fusion operations are often performed to ensure that the modified areas are consistent with their surroundings in terms of color, lighting, and texture, allowing them to blend more seamlessly into the image. Previous methods, when generating datasets, merely performed simple manipulations without further processing. The AIM-DA stage 2 consists of random mask selection, random fusion, and post-processing. random mask selection randomly draws from masks generated in AIM-DA stage 1, using them as masks for fusion. To mimic a more realistic image editing process, we introduce an image blending method and an image harmonization technique in the random fusion stage. Randomly select one of two fusion methods to complete the tampering area fusion. After the fusion stage, the image undergoes overall post-processing. The schematic diagram of random fusion is illustrated in Fig. 2. For image blending, we use a traditional image fusion method based on Poisson image editing, because it is able to naturally transition in most cases without the need for highly accurate target masks. The blending process includes three modes: normal clone, mixed clone, and monochrome transfer. The choice among the three modes is randomized during data augmentation. In order to introduce variability and mimic a wider range of real-world image editing techniques, an end-to-end deep learning method is also employed for automatic image harmonization and enhancement. Such as the Harmonizer [44]. This method introduces a cascade regressor and a dynamic loss strategy for Harmonizer to learn filter arguments more stably and precisely. Following image fusion, post-processing operations are conducted on the image, including simple transforms like rotation, flipping, and transpose.

In summary, the process of image fusion and post-processing begins with the generation of random numbers to determine the chosen image fusion method. Once selected, this method is implemented on a randomly selected tampered area mask to carry out image blending or harmonize operations, thereby enhancing the authenticity of the resulting tampered image. Subsequently, conducting overall post-processing on the tampered image, which may include flipping and rotating, yields

Algorithm 1 Mask Confidence Estimate Algorithm

Input: Prediction mask M_O
Output: Confidence C_O

- 1: Convert the pixel value range of M_O to 0-255, set $C_O = 0$
- 2: Binarize M_O with a threshold of 75 to obtain M_B
- 3: Count the number of connected regions n in M_B
- 4: **if** $n < 10$ **then**
- 5: $C_O = C_O + 1$
- 6: **end if**
- 7: Calculate the percentage of pixels p_{area} in M_O above 0.
- 8: **if** $p_{area} < 0.9$ **then**
- 9: $C_O = C_O + 1$
- 10: **end if**
- 11: Count pixels m_4 between 50 and 170, and m_3 over 15 using histogram statistics.
- 12: Calculate the ratio $p_{diff} = m_4/m_3$
- 13: **if** $p_{area} < 0.1$ AND $p_{diff} < 0.6$ **then**
- 14: $C_O = C_O + 1$
- 15: **else if** $p_{area} \geq 0.1$ AND $p_{diff} < 0.3$ **then**
- 16: $C_O = C_O + 1$
- 17: **end if**
- 18: **return** $C_O = C_O/3$

additional samples while preventing obscuring the created tampering traces.

3.2. Mask confidence estimate based ensemble learning

To enhance the generalization capability of the image manipulation detection algorithm, this paper introduces a MCE-EL module, integrating multiple image manipulation detection models during the inference stage to leverage their strengths for improved overall generalization. Ensemble model selection is based on performance metrics obtained during validation. Each model's contributions are weighted according to their historical accuracy on different manipulation types. This adaptive weighting ensures that the ensemble leverages the strengths of each model effectively. The size of the ensemble can impact performance; a larger ensemble may yield better generalization but at the cost of increased computational complexity. Empirical studies suggest that a balanced ensemble size optimally enhances detection accuracy while maintaining computational efficiency. So, we opt for MVSS-Net [21], CAT-Net [45], and PSCC-Net [19]. They are optimized for efficient feature extraction, leveraging architectures that maintain a balance between performance and computational load. MVSS-Net exhibits the best average performance. CAT-Net, with its unique DCT flow design, effectively detects JPEG compressing artifacts. It adopts HR-Net as the backbone and achieves precise localization for minimal tampering areas. PSCC-Net also utilizes HR-Net for feature extraction and designs a multi-scale feature fusion classification head, excelling in tampering detection classification. Based on the test results, these three models exhibit certain complementary characteristics. Also, our approach is inherently scalable, capable of accommodating additional models or more complex architectures without a linear increase in computational demands. Traditional fusion strategies, such as averaging, taking the maximum, or voting, do not fully capitalize on the strengths of each model. We propose a mask confidence estimate (MCE) algorithm, evaluating the confidence of model outputs to discard low-quality results and combine high-quality ones. It ensures that even with increased model complexity, the system can filter and integrate outputs efficiently. Each model, in predicting the tampered area mask, inevitably incorporates regions of uncertainty. Consequently, the MCE algorithm evaluates the quality of each predicted mask, as detailed in Algorithm 1.

The MCE algorithm considers three factors: first, it eliminates masks with too many scattered areas. If the connected regions in the binarized

mask are more than 10, it is considered a low-quality mask. Secondly, high-quality masks are required to have a tampered area of less than 90%, thereby excluding masks with excessively large tampered regions. Thirdly, the proportion of ambiguous pixels within the entire predicted tampered area is calculated. For small tampered areas less than 10% in size, the proportion of ambiguous pixels must be less than 60%; for tampered areas of other sizes, it must be less than 30% to remove masks that the model predicts with low confidence. The thresholds of 60% and 30% for ambiguous pixels are empirically determined based on extensive validation across multiple datasets. These values are chosen to balance the trade-off between eliminating overly uncertain masks and retaining those with sufficient confidence to be useful. The 60% threshold for smaller tampered areas is set higher to account for the higher likelihood of uncertainty in detecting minimal regions, where model predictions are typically less stable. Conversely, for larger tampered areas, a stricter 30% threshold is applied, as larger regions generally provide more context, allowing the models to make more confident predictions. These values were iteratively tested and validated to achieve optimal performance across diverse scenarios, ensuring that the algorithm remains robust against a wide range of tampering operations. After determining the confidence level, the final prediction mask is obtained by taking the maximum value at each pixel position for masks with a confidence level greater than or equal to 1. If no mask meets the criteria, the final mask will be empty, indicating no tampered area.

In summary, MCE-EL is a confidence-based fusion strategy applied during the inference phase, designed to fully harness the strengths of each model. This approach significantly enhances the overall generalization capability, enabling more effective and accurate analysis of data. By leveraging the MCE algorithm for confidence evaluation, the method effectively filters out low-quality predictions while amalgamating high-quality ones, ensuring that the final ensemble prediction is both reliable and precise.

4. Experiments

4.1. Experimental setup

4.1.1. Training and testing datasets

As shown in Table 1. We utilize 7 training datasets, with CASIAv2 [46] employed for small sample experiments and all 7 datasets for large-scale data training. CASIAv2 [46] includes copy-move and splicing manipulations with post-processing. SP-COCO, CM-COCO, CM-RAISE, CM-C-RAISE, and C-RAISE are derived from [47], containing both single and double JPEG compressed images. RAISE [48] is the source dataset for C-RAISE, comprises about 8000 high-resolution RAW images. For testing, 10 datasets are used. Six of these datasets feature only a single type of manipulation. Columbia [49] and MISD [50] are two splicing manipulation datasets. COVERAGE [51] and CoMo-FoD [52] are copy-move datasets with post-processing. IID-test [53] is a dataset for inpainting, mainly generated by GANs. CocoGlide [54] consists of 512 images generated by diffusion models on the MS-COCO17 [55] validation set and their original images. Four testing datasets include various manipulation types. CASIAv1+[21] features both copy-move and splicing with post-processed [56] tampered areas to blend better into scenes. NIST16 [57] includes all three types of manipulations with diverse image resolutions. DEF12K [21] is sampled from DEFACTo [58], combined with 6000 real images randomly drawn from the MS-COCO14 [55] test set. IMD2020 [59] contains images collected from the internet, without specifying the manipulation type. Each dataset focuses on different aspects to comprehensively test model performance.

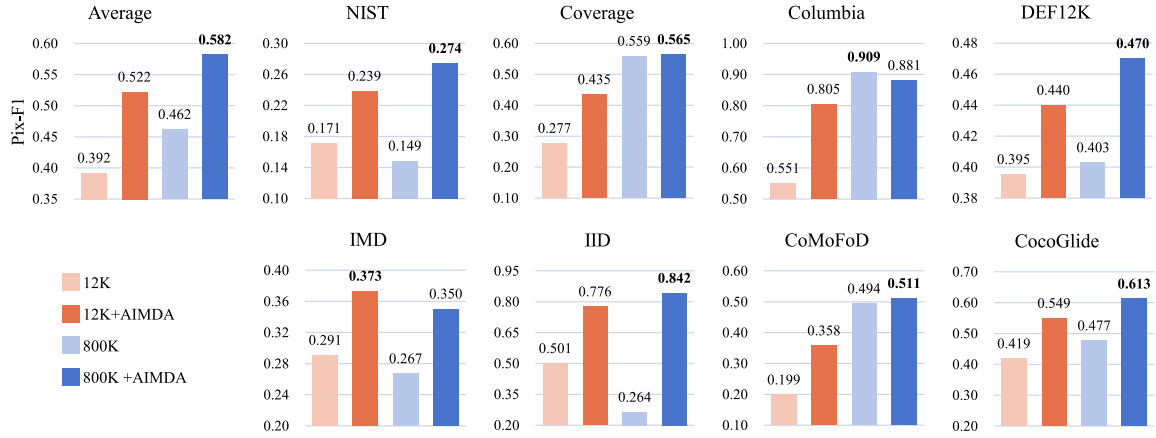


Fig. 3. Performance Comparison of AIM-DA Applied to MVSS-Net on Various Test Sets with Different Training Data Volumes (12K and 800K). Best result per test set is highlighted in bold font.

Table 1

Seven training sets and ten test sets used in our experiments. The symbol — indicates information unavailable. Copy-move, splicing and inpainting are shortened as cpmv, spli and inpa, respectively.

Dataset	Negative	Positive	cpmv	spli	inpa
<i>Training</i>					
CASIAv2 [46]	7,491	5,123	3,295	1,828	0
SP-COCO [47]	0	200,000	0	200,000	0
CM-COCO [47]	0	200,000	200,000	0	0
CM-RAISE [47]	0	200,000	200,000	0	0
CM-C-RAISE [47]	0	200,000	200,000	0	0
C-RAISE [47]	24,462	0	0	0	0
RAISE [48]	8,153	0	0	0	0
<i>Testing</i>					
CASIAv1+ [21]	800	920	459	461	0
Coverage [51]	100	100	100	0	0
Columbia [49]	183	180	0	180	0
IMD2020 [59]	414	2,010	—	—	—
NIST16 [57]	560	564	68	288	208
DEF12K [21]	6,000	6,000	2,000	2,000	2,000
IID-test [53]	0	10,000	0	0	10,000
CoMoFoD [52]	5,000	5,000	5,000	0	0
MISD [50]	0	296	0	296	0
CocoGlide [54]	512	512	0	0	512

4.1.2. Evaluation metrics

Following the convention [21], for manipulation localization, we use pixel-level F1 score as the evaluation criterion, while for manipulation detection, we employ image-level F1 score, which considers both sensitivity and specificity. It is important to note that for authentic images, if the maximum value of the model output mask is 0, then the pixel-level F1 score is set to 1. To closely mirror real-world conditions, the threshold for these metrics is set by default to 0.5, unless otherwise stated.

4.1.3. Implementation details

In AIM-DA stage 1, for inpainting, one of three methods is randomly chosen: INPAINT_TELEA [41], INPAINT_NS [42], or the deep learning approach GC [43]. In AIM-DA stage 2, we adopt Poisson image editing [60] for image blending and choose the Harmonizer [44] method for image harmonization and enhancement.

Models are implemented using the PyTorch framework and trained on two NVIDIA A800 PCIe 80 GB GPUs, with one dedicated to model training and the other to data augmentation. For MVSS-Net, the backbone network is initialized with ImageNet pre-trained parameters. Adam is used as the optimizer. The learning rate undergoes a warm-up over one epoch, linearly increasing from 10^{-7} to 10^{-4} , followed by a cosine decay back to 10^{-7} . The model is trained for 50 epochs with a

batch size of 28. For CAT-Net, the DCT branch is initialized with the weights pre-trained by the authors, and the RGB branch is initialized with ImageNet pre-trained parameters. The optimizer is SGD, with an initial learning rate set at 0.005, decaying to 10^{-7} over 200 epochs. The batch size is set to 12. PSCC-Net's backbone network is also initialized with ImageNet pre-trained parameters. The optimizer is Adam, with the learning rate starting at 2×10^{-4} and step-wise decreasing to 1.25×10^{-5} , halving every 5 epochs. The model is trained for 25 epochs with a batch size of 16.

4.2. Quantitative comparisons

To verify the effectiveness of the proposed method, we conduct experiments in two parts: data augmentation and model ensembling. The effectiveness of the proposed AIM-DA method is evaluated using MVSS-Net as an example. We first investigate how different data augmentation tampering ratios impact model performance. Following this initial assessment, the effectiveness of AIM-DA is tested across varying data volumes, specifically comparing scenarios with 12K and 800K images. Lastly, a comparative analysis is performed, juxtaposing the improvements attributable to AIM-DA and MCE-EL against other data augmentation and ensemble methodologies.

In order to further verify the effectiveness of data enhancement, we conducted model performance experiments with different data enhancement tampering ratios and model performance experiments with different data enhancement post-processing ratios as shown in Fig. 4. The proposed AIM-DA method is dissected into tampered image generation and fusion post-processing for evaluation. Fig. 4(a) illustrates model performance across varying tampered image generation ratios, where a ratio of 0, indicating the absence of data augmentation, correlates with inferior model performance. The model's performance peaks at a ratio of 0.9, at which juncture the majority of images within an epoch are tampered with. The adoption of a dynamically weighted loss function during training mitigates the impact of data imbalance on model efficacy. Elevating the ratio to 1, which entails data augmentation across all samples, precipitates a notable decline in performance, attributed to the exclusion of authentic images from training, resulting in heightened sensitivity, reduced specificity, and an increased propensity for false positives. Fig. 4(b), based on the optimal tampered image generation ratio, assesses model performance across different fusion post-processing ratios, identifying 0.2 as the optimal threshold. Ratios exceeding this value predispose the model to overfit to post-processing techniques, detrimentally affecting its generalization.

Fig. 3 compares the effectiveness of the AIM-DA data augmentation method across two different data volumes: 12K and 800K. The 12K dataset exclusively comprises the CASIAv2 training set, whereas the

Table 2

Performance on pixel-level manipulation localization. Performance metric: pixel-level F1. The best two results per test set are highlighted in bold font, with the top result underlined. All models are trained on CASIAv2, except for those marked with a star (*), the training data of which contains other datasets or not publicly accessible data.

Method	Pixel-level F1										
	CASIAv1+	Columbia	COVER	NIST	IMD	IID	DEF-12K	CoMoFoD	MISD	CocoGlide	MEAN
PSCC-Net [19]	0.579	0.241	0.097	0.176	0.229	0.291	0.057	0.051	0.662	0.098	0.248
CAT-Net*[45]	0.665	0.212	0.157	0.267	0.219	0.721	0.708	0.026	0.451	0.178	0.360
MVSS-Net [21]	0.630	0.551	0.277	0.171	0.291	0.501	0.395	0.199	0.724	0.419	0.416
<i>Data augmentation:</i>											
MVSS+CutOut [23]	0.624	0.586	0.237	0.206	0.323	0.394	0.341	0.156	0.752	0.416	0.403
MVSS+GridMask [24]	0.616	0.526	0.240	0.193	0.292	0.446	0.387	0.266	0.729	0.390	0.408
MVSS+RandAug [25]	0.618	0.512	0.226	0.191	0.313	0.233	0.219	0.169	0.760	0.182	0.342
MVSS+MixStyle [30]	0.624	0.497	0.261	0.133	0.303	0.557	0.363	0.237	0.744	0.298	0.402
PSCC+AIM-DA	0.576	0.430	0.168	0.185	0.198	0.695	0.197	0.083	0.565	0.324	0.342
CAT*+AIM-DA	0.693	0.216	0.153	0.323	0.210	0.762	0.784	0.023	0.457	0.183	0.380
MVSS+AIMDA	0.631	0.805	0.435	<u>0.239</u>	0.373	0.776	<u>0.440</u>	0.358	0.723	0.549	0.533
MVSS*+AIM-DA	0.708	0.881	0.565	0.274	0.350	0.842	0.470	0.511	0.731	0.613	0.595
<i>Ensemble:</i>											
Intersection	0.606	0.626	0.437	0.092	0.219	0.791	0.525	0.359	0.624	0.546	0.483
Union	0.658	0.219	0.139	0.302	0.282	0.623	0.424	0.024	0.460	0.189	0.332
Min	0.606	0.626	0.437	0.092	0.219	0.791	0.525	0.359	0.624	0.546	0.483
Max	0.658	0.219	0.139	0.302	0.282	0.623	0.424	0.024	0.460	0.189	0.332
HardVote	0.632	0.616	0.228	0.254	0.301	0.832	0.527	0.100	0.654	0.338	0.448
SoftVote	0.632	0.621	0.237	0.269	0.291	0.834	0.556	0.102	0.664	0.341	0.455
MCE-EL	0.670	0.805	0.443	0.326	0.380	0.773	0.699	0.358	0.731	0.548	0.573

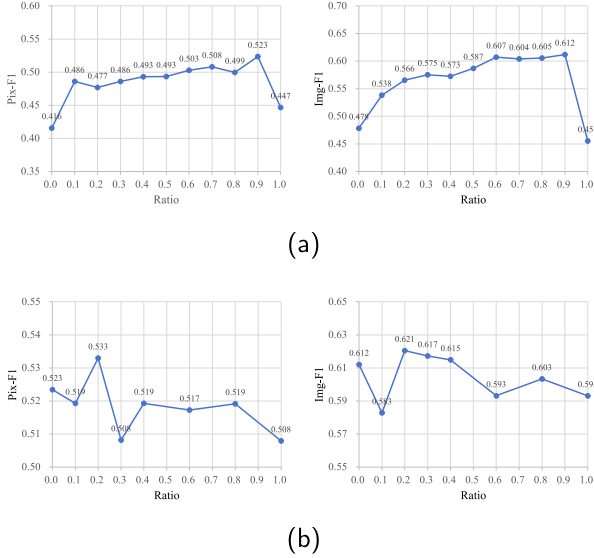


Fig. 4. Pixel-level and image-level F1 of MVSS-Net under different AIM-DA ratios. (a) Tampered image generation. (b) Fusion & post-processing.

800K dataset includes data from all training sets listed in Table 1. The results indicate that the proposed AIM-DA method effectively enhances model performance across varying training data volumes, achieving an average performance increase of 13% on the 12K dataset and 12% on the 800K dataset. Specifically, on the COVERAGE and CoMoFoD copy-move test sets, greater improvements are observed for the smaller dataset, with lesser gains for the larger dataset, attributable to the 800K dataset's extensive inclusion of copy-move tampered images. For the IID inpainting test set, the performance without data augmentation is lower for the larger dataset compared to the smaller one, likely due to a lack of inpainting images in the training set and a predominance of splicing and copy-move manipulations, leading to decreased performance; however, introducing data augmentation results in a significant performance boost. When using the 800K dataset trained with AIM-DA for the Columbia dataset, a slight performance decrease is observed, potentially due to the dataset's characteristic simple, large-area, arbitrary splicing. The tampering masks utilized by

AIM-DA predominantly target smaller areas, and the introduction of numerous complex tampered samples may lead the model to overlook these simplistic, extensive manipulations.

In Table 2, we conduct comparative experiments between the proposed AIM-DA and MCE-EL against existing state-of-the-art approaches. Starting with a comparison against popular data augmentation methods. We observe that model performance does not improve and even declines after applying classic dropout-type data augmentations such as CutOut [23] or GridMask [24]. This is primarily because these methods fill certain image regions with zeros, which could be misinterpreted as a form of rare tampering, potentially leading to model overfitting. RandAug [25], which randomly applies 16 types of enhancements including auto-contrast and sharpening through two adjustable hyperparameters, showed unsatisfactory results for image manipulation detection tasks, likely because the augmented data obscured original tampering traces, hindering the model's ability to learn more generalized tampering patterns. MixStyle [30], which introduces a MixStyle layer into the model to blend image style features and improve generalization, is applied as recommended by its authors at the third and fourth stages of the model, yielding moderate results. This outcome suggests that feature-level mixing may not be well-suited for image manipulation detection tasks. The AIM-DA method, specifically designed for image manipulation detection tasks, generates a broader range of tampering traces along with their corresponding tampered area masks and tampered edge masks, effectively enhancing model performance.

The lower sections of Tables 2 and 3 present the pixel-level and image-level F1 scores when ensembling MVSS-Net, PSCC-Net, and CAT-Net trained with AIM-DA using various ensemble methods operational at the inference stage. These ensemble strategies include binary thresholding the output masks of the three models at a 0.5 threshold and taking either the intersection or union; choosing the minimum or maximum prediction probability at each pixel position; voting at each pixel position after Binarization, where a pixel is deemed tampered if at least two models agree, setting the final prediction to 1, otherwise 0; averaging the prediction probabilities at each pixel position for a soft vote; and, in the final row, fusion using the proposed MCE algorithm. Traditional ensemble methods appear to inadequately capitalize on the strengths of individual models, for instance, CAT-Net shows exceptional performance on the DEF12K dataset with a pixel-level F1 of 0.784, yet its performance notably declines after integration. Employing the proposed confidence-based ensemble method, where uncertain masks are discarded, mitigates this performance decrease. Overall, the MCE-EL strategy leads to an average pixel-level F1 improvement of 4.0% and

Table 3

Performance on image-level manipulation detection. Decision threshold: 0.5. NIST, IID, and MISD are excluded as they have no authentic image. All models are trained on CASIAv2, except for those marked with a star (*), the training data of which contains other datasets or not publicly accessible data.

Method	Image-level F1							
	CASIAv1+	Columbia	COVER	IMD	DEF-12K	CoMoFoD	CocoGlide	MEAN
PSCC-Net [19]	0.505	0.170	0.020	0.334	0.127	0.000	0.150	0.187
CAT-Net* [45]	0.697	0.000	0.058	0.494	0.882	0.000	0.004	0.305
MVSS-Net [21]	0.579	0.788	0.493	0.632	0.537	0.350	0.559	0.563
<i>Data augmentation:</i>								
MVSS+CutOut [23]	0.535	0.818	0.408	0.654	0.534	0.261	0.439	0.521
MVSS+GridMask [24]	0.551	0.696	0.445	0.630	0.509	0.470	0.480	0.540
MVSS+RandAug [25]	0.664	0.723	0.338	0.580	0.417	0.259	0.285	0.467
MVSS+MixStyle [30]	0.540	0.749	0.464	0.661	0.532	0.459	0.521	0.561
PSCC+AIM-DA	0.494	0.738	0.462	0.572	0.361	0.480	0.643	0.536
CAT*+AIM-DA	0.735	0.000	0.058	0.540	0.959	0.000	0.000	0.327
MVSS+AIM-DA	0.531	0.918	0.594	0.688	0.459	0.541	0.613	0.621
MVSS*+AIM-DA	0.682	0.972	0.667	0.651	0.469	0.361	0.715	0.645
<i>Ensemble:</i>								
Intersection	0.494	0.916	0.597	0.272	0.176	0.509	0.578	0.506
Union	0.788	0.000	0.020	0.248	0.425	0.000	0.000	0.212
Min	0.494	0.916	0.597	0.272	0.176	0.509	0.578	0.506
Max	0.788	0.000	0.020	0.248	0.425	0.000	0.000	0.212
HardVote	0.552	0.683	0.113	0.632	0.464	0.130	0.330	0.415
SoftVote	0.551	0.677	0.113	0.594	0.474	0.130	0.303	0.406
MCE-EL	0.637	0.918	0.597	0.711	0.808	0.541	0.610	0.689

Table 4

Ablation study of AIM-DA. Training: CASIAv2. Test: all ten test datasets. Pixel-level F1 (Pix-F1), image-level F1 (Img-F1), combine-F1 (Com-F1). The best per column is highlighted in bold.

No.	Setup						Pix-F1	Img-F1	Com-F1
	Cp	Sp	Ip	Ct	Ps	Bl			
#0:							0.416	0.563	0.490
#1:	✓						0.440	0.540	0.490
#2:		✓					0.433	0.529	0.481
#3:			✓				0.463	0.590	0.527
#4:	✓	✓	✓				0.477	0.566	0.522
#5:	✓	✓	✓	✓			0.455	0.608	0.532
#6:	✓	✓	✓		✓		0.481	0.559	0.520
#7:	✓	✓	✓			✓	0.472	0.485	0.479
#8:	✓	✓	✓		✓	✓	0.501	0.612	0.557
#9:	✓	✓	✓	✓	✓	✓	0.471	0.588	0.530

an average image-level F1 increase of 6.8% compared to the standalone models, affirming the effectiveness of the proposed MCE-EL.

4.3. Qualitative comparisons

Fig. 5 shows the visual comparison of forgery localization quality of the MCE-EL integrated inference results of MVSS-Net, PSCC-Net, and CAT-Net, trained using the proposed data augmentation method, against SOTA methods through three types of images. The examples presented include both tampered and original genuine images to showcase the quality of forgery localization. Our method demonstrates superior and comprehensive localization of tampered regions. For instance, Fig. 5(b) in the first row featuring a copy-move tampered image, most models incorrectly identify only the duplicated signposts on the grass as tampered areas, neglecting the extensive copy-move

manipulation throughout the surrounding grass area. Our approach more accurately distinguishes between genuine and manipulated areas, significantly reducing false alarms, as evidenced in the handling of the original image of the same scene. The third-row image, inpainting tampered image from the CocoGlide dataset, showcases our method's ability to precisely detect both tampered regions, contrasting with other approaches that either miss or falsely detect manipulations. For image restoration of irregular areas such as Fig. 5(c) the first and second rows, the method in this paper can more accurately identify the tampered areas. In authentic images, our approach preserves the characteristic low false alarm rate of MVSS-Net, unlike other methods that tend to mistakenly label untampered areas as manipulated.

4.4. Ablation studies

To assess the impact and efficacy of each component within the proposed AIM-DA method, ablation studies are conducted using MVSS-Net trained on the CASIAv2 training dataset. The results are presented in Table 4, where six operations are evaluated: copy-move tampering (Cp), image splicing tampering (Sp), inpainting tampering (Ip), random lighting contrast color transformation (Ct), Poisson image editing (Ps), and image blending (Bl). The pixel-level F1 scores are calculated from ten out-of-domain test datasets, with the rightmost column (Com-F1) showing the average of Pix-F1 and Img-F1. Row #0 serves as the baseline, without any data augmentation.

Compared to the baseline (#0), the inclusion of any of the operations Cp, Sp, or Ip results in improved pixel-level F1 scores. Notably, after incorporating the Ip operation, which is missing from the dataset, a significant performance increase is observed in #3 on the IID and CocoGlide, two inpainting test sets. The model achieves better performance when all three tampering operations are used, as indicated in row #4.

Table 5

Pixel-level F1 performance (fixed threshold) on datasets uploaded on Without (w/o), Facebook (Fb), WhatsApp (Wa), Weibo (Wb), WeChat (Wc). The best result per column is underlined, and the second-best result is in bold.

Method	CASIAv1+					Columbia					NIST16					AVG				
	w/o	Fb	Wa	Wb	Wc	w/o	Fb	Wa	Wb	Wc	w/o	Fb	Wa	Wb	Wc	w/o	Fb	Wa	Wb	Wc
IF-OSN	0.5091	0.4638	0.4775	0.4658	0.4053	0.7127	0.7164	0.7274	0.7261	0.7322	0.3309	0.3258	0.3134	0.2895	0.2805	0.5176	0.5020	0.5061	0.4938	0.4727
MVSS-Net	0.4513	0.3874	0.3588	0.4035	0.2483	0.6841	0.6909	0.6852	0.6890	0.6900	0.2945	0.2642	0.1651	0.2504	0.2113	0.4766	0.4475	0.4030	0.4476	0.3832
CAT-Net	0.7066	0.6301	0.4215	0.4246	0.1387	0.7933	0.9175	0.9208	0.9212	0.8481	0.2846	0.1505	0.2009	0.2080	0.1913	0.5948	0.5660	0.5144	0.5179	0.3927
PSCC-Net	0.6268	0.5837	0.5736	0.5057	0.4894	0.8661	0.8381	0.8422	0.4059	0.8338	0.2981	0.2860	0.2947	0.1592	0.2679	0.5970	0.5693	0.5702	0.3569	0.5304
Ensemble (ours)	0.6316	0.5483	0.4822	0.5078	0.3060	0.9007	0.9122	0.9087	0.8924	0.8964	0.4231	0.3016	0.2794	0.2856	0.2914	0.6518	0.5873	0.5567	0.5619	0.4979

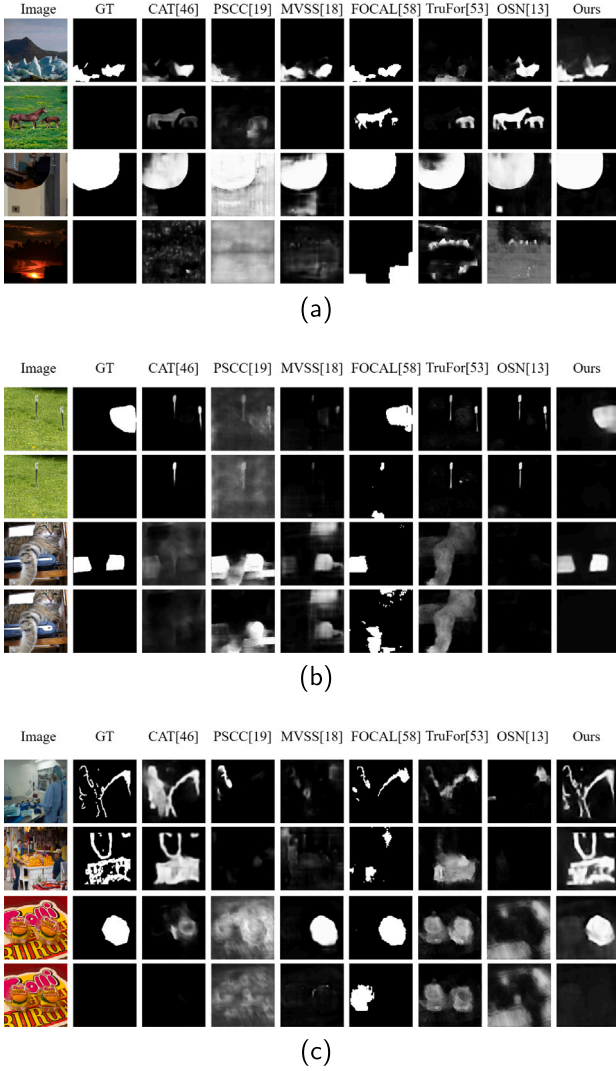


Fig. 5. Qualitative results on different forged images. For each row, the images from left to right are forgery, ground-truth forgery mask, detection results generated by CAT-Net [45], PSCC-Net [19], MVSS-Net [21], IF-OSN [17], FOCAL [61] and TruFor [54], respectively. “Ours” refers to the ensemble results of PSCC-Net, CAT-Net, and MVSS-Net using GES.

Further analysis on the effectiveness of fusion post-processing reveals that while the addition of Background Inpainting (BI) alone in row#7 decreases performance, its combination with Post-Processing (PS) in row#8 yields an improvement. This can be attributed to the fact that PS enhances the visual coherence of the manipulated images, which may counterbalance the artifacts introduced by BI alone. The synergy between PS and BI creates a more holistic approach to image manipulation, allowing the model to learn more effectively from the

resultant images. The addition of Ct leads to a decrease in performance, similar to the results observed when testing RandAug data augmentation in Table 2. The Ct operation likely hampers the model’s ability to recognize tampering traces, possibly due to overfitting to unrealistic brightness and contrast settings during training. Consequently, the final AIM-DA configuration is chosen as per the settings in #8.

The results in Table 4 demonstrate the effectiveness of AIM-DA’s cumulative tampering approach. The significant performance improvements (Pix-F1, Img-F1) across diverse tampering types show that AIM-DA contributes to robust detection [62] across unknown tampering operations. These experiments provide clear evidence of the enhanced generalization and efficacy in open domain tampering detection. (see Fig. 6).

4.5. Additional robustness analysis

In this subsection. We include additional experiments to show the ability of our method to be robust to different forms of degradations and compare them with those obtained by the top performers [17, 19, 21, 45]. We apply the following transformations on the CASIAv1+ dataset: JPEG compression (varying the quality level), gaussian blur (varying the kernel size), gaussian noise (varying the standard deviation) and Resize (varying the resize factor). The results are shown in Fig. 3. We can observe that our method is more robust than the state-of-the-art irrespective of the type of degradation.

We also check robustness to other social media networks, beyond those already considered in the main paper, i.e. Facebook and Whatsapp. More specifically, we use the whole dataset proposed in [17], CASIAv1+ [21], Columbia [49], and NIST16 [57], were also uploaded on Weibo and WeChat. Results are presented in Table 5 and show a consistent gain over all the different datasets and social platforms. Notably, CAT-Net demonstrates superior performance on the Columbia dataset, achieving the best results in all evaluation metrics. On average however, Our method is competitive and advantageous across various test datasets.

5. Conclusion

In this paper, we develop a simple yet effective GES for image manipulation detection algorithms, which aims to conveniently enhance the generalization of existing models through data augmentation and model ensembling. Specifically, we build the AIM-DA module, which consists of several cumulative tampering operations intended to mimic realistic scenarios of image manipulation. This approach significantly enriches the dataset with a wide range of tampering traces. The MCE-EL module integrates outputs from various image manipulation detection models, enabling a multi-perspective image examination that capitalizes on the unique strengths of each algorithm to boost image manipulation detection accuracy and generalization. Experimental evaluation on several popular cross-domain benchmarks verifies the effectiveness of our proposed method. However, the method may still show reduced performance in highly simplistic, large-area tampering operations (e.g., Columbia dataset), and future work may focus on addressing this limitation by incorporating larger-scale, simpler tampering traces in the training process.

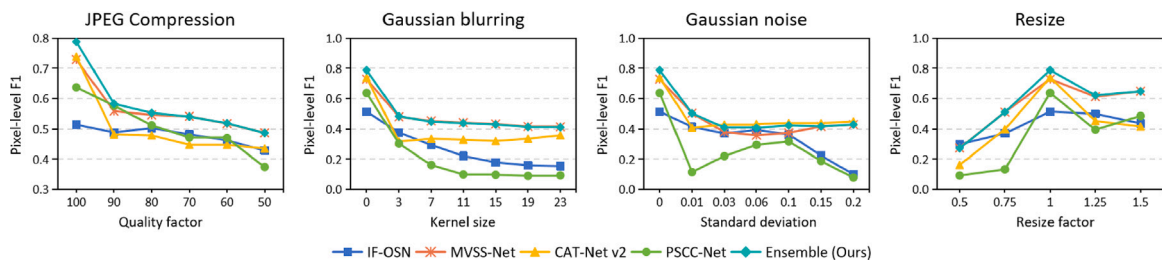


Fig. 6. Robustness analysis [62] against different processing operations on CASIAv1+. Pixel-level F1 performance (best threshold) is shown.

CRediT authorship contribution statement

H. Cheng: Methodology, Conceptualization. **L. Niu:** Methodology. **Z. Zhang:** Software, Methodology. **L. Ye:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Zan Gao, Chao Sun, Zhiyong Cheng, Weili Guan, Anan Liu, Meng Wang, Tbnnet: A two-stream boundary-aware network for generic image manipulation localization, *IEEE Trans. Knowl. Data Eng.* 35 (7) (2023) 7541–7556.
- [2] R. Salloum, Y. Ren, C.-C. Jay Kuo, Image splicing localization using a multi-task fully convolutional network (MFCN), *J. Vis. Commun. Image Represent.* 51 (2018) 201–209.
- [3] J.-L. Zhong, C.-M. Pun, An end-to-end dense-InceptionNet for image copy-move forgery detection, *IEEE Trans. Inf. Forensics Secur.* 15 (2020) 2134–2146.
- [4] J. Bunk, et al., Detection and localization of image forgeries using resampling features and deep learning, in: *IEEE Conf. on Comput. Vis. Pattern Recognit. Workshops, CVPRW, 2017*, pp. 1881–1889.
- [5] P. Ferrara, T. Bianchi, A. De Rosa, A. Piva, Image forgery localization via fine-grained analysis of CFA artifacts, *IEEE Trans. Inf. Forensics Secur.* 7 (5) (2012) 1566–1577.
- [6] J. Park, D. Cho, W. Ahn, H.-K. Lee, Double JPEG detection in mixed JPEG quality factors using deep convolutional neural network, in: *Proc. Eur. Conf. Comput. Vis., ECCV, 2018*, pp. 636–652.
- [7] B. Mahdian, S. Saic, Using noise inconsistencies for blind image forensics, *Image Vis. Comput.* 27 (10) (2009) 1497–1503.
- [8] H. Li, J. Huang, Localization of deep inpainting using high-pass fully convolutional network, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis., ICCV, 2019*, pp. 8301–8310.
- [9] P. Zhou, X. Han, V.I. Morariu, L.S. Davis, Learning rich features for image manipulation detection, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2018*, pp. 1053–1061.
- [10] J. Hao, Z. Zhang, S. Yang, D. Xie, S. Pu, TransForensics: Image forgery localization with dense self-attention, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis., ICCV, 2021*, pp. 15055–15064.
- [11] Y. Rao, J. Ni, Self-supervised domain adaptation for forgery localization of JPEG compressed images, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis., ICCV, 2021*, pp. 15014–15023.
- [12] P. Zhou, et al., Generate, segment, and refine: Towards generic manipulation segmentation, in: *Proc. AAAI Conf. Artif. Intell., vol. 34, 2020*, no. 07, Art. no. 07.
- [13] A. Kirillov, et al., Segment anything, 2023, [arXiv:2304.02643](https://arxiv.org/abs/2304.02643).
- [14] Y. Wu, W. AbdAlmageed, P. Natarajan, ManTra-net: Manipulation tracing network for detection and localization of image forgeries with anomalous features, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2019*, pp. 9543–9552.
- [15] J. Fridrich, J. Kodovsky, Rich models for steganalysis of digital images, *IEEE Trans. Inf. Forensics Secur.* 7 (3) (2012) 868–882.
- [16] B. Bayar, M.C. Stamm, Constrained convolutional neural networks: A new approach towards general purpose image manipulation detection, *IEEE Trans. Inf. Forensics Secur.* 13 (11) (2018) 2691–2706.
- [17] H. Wu, J. Zhou, J. Tian, J. Liu, Robust image forgery detection over online social network shared images, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2022*, pp. 13440–13449.
- [18] X. Hu, Z. Zhang, Z. Jiang, S. Chaudhuri, Z. Yang, R. Nevatia, SPAN: Spatial pyramid attention network for image manipulation localization, in: *Proc. Eur. Conf. Comput. Vis., ECCV, 2020*, pp. 312–328.
- [19] X. Liu, Y. Liu, J. Chen, X. Liu, PSCC-net: Progressive spatio-channel correlation network for image manipulation detection and localization, *IEEE Trans. Circuits Syst. Video Technol.* (2022) 1–1.
- [20] Zan Gao, Shenao Chen, Yangyang Guo, Weili Guan, Jie Nie, Anan Liu, Generic image manipulation localization through the lens of multi-scale spatial inconsistency, in: *Proceedings of the 30th ACM International Conference on Multimedia, MM'22, Lisbon, Portugal, 2022*.
- [21] C. Dong, X. Chen, R. Hu, J. Cao, X. Li, MVSS-net: Multi-view multi-scale supervised networks for image manipulation detection, *IEEE Trans. Pattern Anal. Mach. Intell.* (2022) 1–14.
- [22] D. Xu, X. Shen, Z. Shi, N. Ta, Semantic-agnostic progressive subtractive network for image manipulation detection and localization, *Neurocomputing* 543 (2023) 126263.
- [23] T. DeVries, G.W. Taylor, Improved regularization of convolutional neural networks with cutout, 2017, [arXiv:1708.04552](https://arxiv.org/abs/1708.04552).
- [24] P. Chen, S. Liu, H. Zhao, X. Wang, J. Jia, GridMask data augmentation, 2024, [arXiv:2001.04086](https://arxiv.org/abs/2001.04086).
- [25] E.D. Cubuk, B. Zoph, J. Shlens, Q.V. Le, RandAugment: Practical automated data augmentation with a reduced search space, 2019, [arXiv:1909.13719](https://arxiv.org/abs/1909.13719).
- [26] A. Prakash, et al., Structured domain randomization: Bridging the reality gap by context-aware synthetic data, in: *Proc. Int. Conf. Rob. Autom., ICRA, 2019*, pp. 7249–7255.
- [27] K. Zhou, Y. Yang, T. Hospedales, T. Xiang, Deep domain-adversarial image generation for domain generalisation, in: *Proc. AAAI Conf. Artif. Intell., vol. 34 (2020) no. 07, Art. no. 07*.
- [28] Y. Zhang, D. Zhou, B. Hooi, K. Wang, J. Feng, Expanding small-scale datasets with guided imagination, 2023, [arXiv:2211.13976](https://arxiv.org/abs/2211.13976).
- [29] H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, Mixup: Beyond empirical risk minimization, 2018, [arXiv:1710.09412](https://arxiv.org/abs/1710.09412).
- [30] K. Zhou, Y. Yang, Y. Qiao, T. Xiang, Domain generalization with MixStyle, 2021, [arXiv:2104.02008](https://arxiv.org/abs/2104.02008).
- [31] P. Zhuang, H. Li, S. Tan, B. Li, J. Huang, Image tampering localization using a dense fully convolutional network, *IEEE Trans. Inf. Forensics Secur.* 16 (2021) 2986–2999.
- [32] J. Wang, et al., Generalizing to unseen domains: A survey on domain generalization, *IEEE Trans. Knowl. Data Eng.* (2022) 1–1.
- [33] L. Breiman, Bagging predictors, *Mach. Learn.* 24 (2) (1996) 123–140.
- [34] Z. Xu, W. Li, L. Niu, D. Xu, Exploiting low-rank structure from latent domains for domain generalization, in: *Proc. Eur. Conf. Comput. Vis., ECCV, 2014*, pp. 628–643.
- [35] K. Zhou, Y. Yang, Y. Qiao, T. Xiang, Domain adaptive ensemble learning, *IEEE Trans. Image Process.* 30 (2021) 8008–8018.
- [36] M. Mancini, S.R. Bulò, B. Caputo, E. Ricci, Best sources forward: Domain generalization through source-specific nets, in: *Proc. 25th IEEE Int. Conf. Image Process., ICIP, 2018*, pp. 1353–1357.
- [37] S. Wang, L. Yu, K. Li, X. Yang, C.-W. Fu, P.-A. Heng, DoFE: Domain-oriented feature embedding for generalizable fundus image segmentation on unseen datasets, *IEEE Trans. Med. Imaging* 39 (12) (2020) 4237–4248.
- [38] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, A.G. Wilson, Averaging weights leads to wider optima and better generalization, 2019, [arXiv:1803.05407](https://arxiv.org/abs/1803.05407).
- [39] T. Sämman, A.M. Hammam, A. Bursuc, C. Stiller, H.-M. Groß, Improving predictive performance and calibration by weight fusion in semantic segmentation, 2022, [arXiv:2207.11211](https://arxiv.org/abs/2207.11211).
- [40] G. Liu, F.A. Reda, K.J. Shih, T.-C. Wang, A. Tao, B. Catanzaro, Image inpainting for irregular holes using partial convolutions, in: *Proc. Eur. Conf. Comput. Vis., ECCV, 2018*, pp. 85–100.
- [41] A. Telea, An image inpainting technique based on the fast marching method, *J. Graph. Tools* 9 (1) (2004) 23–34.
- [42] M. Bertalmio, A.L. Bertozzi, G. Sapiro, Navier–Stokes, fluid dynamics, and image and video inpainting, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit., CVPR, 2001*, pp. 1–355–I–362.
- [43] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, T.S. Huang, Free-form image inpainting with gated convolution, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis., ICCV, 2019*, pp. 4471–4480.
- [44] Z. Ke, C. Sun, L. Zhu, K. Xu, R.W.H. Lau, Harmonizer: Learning to perform white-box image and video harmonization, in: *Proc. Eur. Conf. Comput. Vis., ECCV, 2022*, pp. 690–706.
- [45] M.-J. Kwon, S.-H. Nam, I.-J. Yu, H.-K. Lee, C. Kim, Learning JPEG compression artifacts for image manipulation detection and localization, *Int. J. Comput. Vis.* 130 (8) (2022) 1875–1895.
- [46] J. Dong, W. Wang, T. Tan, CASIA image tampering detection evaluation database, in: *IEEE China Summit Int. Conf. Signal Inf. Process., ChinaSIP, 2013*, pp. 422–426.
- [47] M.-J. Kwon, I.-J. Yu, S.-H. Nam, H.-K. Lee, CAT-net: Compression artifact tracing network for detection and localization of image splicing, in: *Proc. IEEE Winter Conf. Appl. Comput. Vis., WACV, 2021*, pp. 375–384.
- [48] D.-T. Dang-Nguyen, C. Pasquini, V. Conotter, G. Boato, RAISE: a raw images dataset for digital image forensics, in: *Proc. 6th ACM Multimedia Syst. Conf., MMSys, 2015*, pp. 219–224.

- [49] Y. Hsu, S. Chang, Detecting image splicing using geometry invariants and camera characteristics consistency, in: Proc. IEEE Int. Conf. Multimedia Expo, ICME, 2006, pp. 549–552.
- [50] K.D. Kadam, S. Ahirrao, K. Kotecha, Multiple image splicing dataset (MISD): A dataset for multiple splicing, *Data* 6 (10) (2021) 102.
- [51] B. Wen, Y. Zhu, R. Subramanian, T.-T. Ng, X. Shen, S. Winkler, COVERAGE — A novel database for copy-move forgery detection, in: Proc. IEEE Int. Conf. Image Process., ICIP, 2016, pp. 161–165.
- [52] D. Tralic, I. Zupancic, S. Grgic, M. Grgic, CoMoFoD — New database for copy-move forgery detection, in: Proc. Int. Symp. Electron. Mar., ELMAR, 2013, pp. 49–54.
- [53] H. Wu, J. Zhou, IID-net: Image inpainting detection network via neural architecture search and attention, *IEEE Trans. Circuits Syst. Video Technol.* 32 (3) (2022) 1172–1185.
- [54] F. Guillaro, D. Cozzolino, A. Sud, N. Dufour, L. Verdoliva, TruFor: Leveraging all-round clues for trustworthy image forgery detection and localization, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., CVPR, 2023, pp. 20606–20615.
- [55] T.-Y. Lin, et al., Microsoft COCO: Common objects in context, in: Proc. Eur. Conf. Comput. Vis., ECCV, 2014, pp. 740–755.
- [56] Jiaxin Chen, Xin Liao, Wei Wang, Zhenxing Qian, Zheng Qin, Yaonan Wang, SNIS: A signal noise separation-based network for post-processed image forgery detection, *IEEE Trans. Circuits Syst. Video Technol.* 33 (2) (2023) 935–951.
- [57] H. Guan, et al., MFC datasets: Large-scale benchmark datasets for media forensic challenge evaluation, in: Proc. IEEE Winter Conf. Appl. Comput. Vis. Workshops, WACVW, 2019, pp. 63–72.
- [58] G. Mahfoudi, B. Tajini, F. Retraint, F. Morain-Nicolier, J.L. Dugelay, M. Pic, DEFACTO: Image and face manipulation dataset, in: Proc. 27th Eur. Signal Process. Conf., EUSIPCO, 2019, pp. 1–5.
- [59] A. Novozamsky, B. Mahdian, S. Saic, IMD2020: A large-scale annotated dataset tailored for detecting manipulated images, in: Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops, WACVW, 2020, pp. 71–80.
- [60] P. Pérez, M. Gangnet, A. Blake, Poisson image editing, in: ACM SIGGRAPH Pap., 2003, pp. 313–318.
- [61] H. Wu, Y. Chen, J. Zhou, Rethinking image forgery detection via contrastive learning and unsupervised clustering, 2023, [arXiv:2308.09307](https://arxiv.org/abs/2308.09307).
- [62] Xin Liao, Kaide Li, Xinshan Zhu, K.J. Ray Liu, Robust detection of image operator chain with two-stream convolutional neural network, *IEEE J. Sel. Top. Signal Process.* 14 (5) (2020) 955–968.